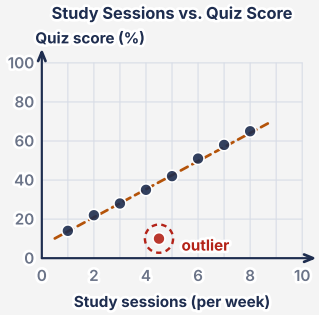
📝 Unit 7 Practice 2 — Data Analysis

Complete independently · NC.8.SP.1–4 · Show all work

Name _____ Date _____ Period _____

Remember: Describe a scatter plot: Up or down? Tight or spread out? Any outliers?

Question 1



Looking at the **non-outlier** points, what correlation is shown?

- ○ Negative correlation
- ○ Positive correlation
- ○ No correlation
- ○ Perfect correlation

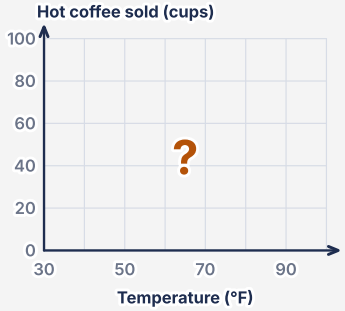
Question 2



This scatterplot shows athletes' skill level as weeks without practice increase. Describe the full correlation.

- ○ Weak positive correlation — practice doesn't matter
- ○ Strong negative — the longer they skip practice, the lower their skill
- ○ No correlation — the points are scattered
- ○ Moderate positive — skill improves with rest

Question 3



In a scatterplot for "temperature vs. hot coffee sales," which type of correlation would you expect?

- ○ Strong positive
- ○ Strong negative
- ○ No correlation
- ○ Perfect positive

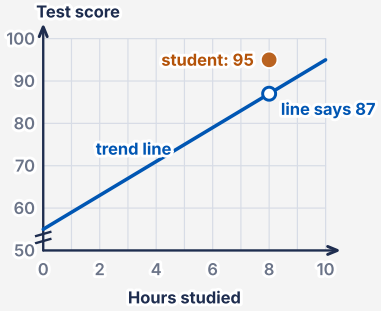
Question 4



A scatterplot shows "number of hours of daylight vs. hours of sleep." Which correlation is most likely?

- ○ Strong positive
- ○ Weak to no correlation
- ○ Strong negative
- ○ Perfect negative

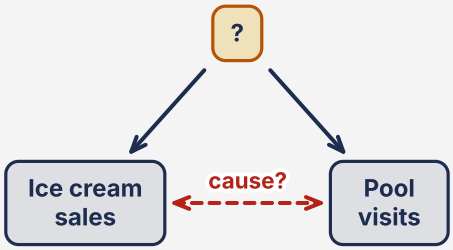
Question 5



Student scores a 95 on a test after studying 8 hours. The trend line predicts 87 points at 8 hours. This student's score above the line makes it:

- ○ A point above the trend line (the line under-predicts this student)
- ○ An outlier below the trend line
- ○ The y-intercept
- ○ The maximum score

Question 6

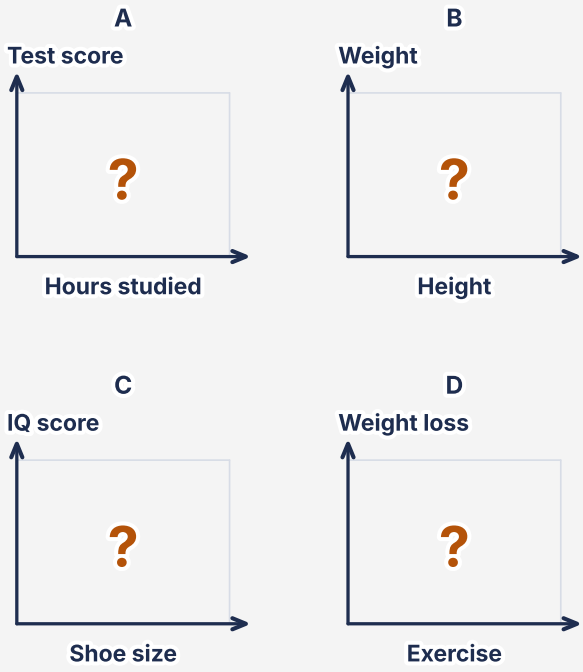


Both go up in summer. Is one causing the other, or is something else behind both?

Ice cream sales and pool visits both increase in summer. A student says "ice cream causes people to visit pools." What concept explains the real relationship?

- ○ Causation — ice cream makes people want to swim
- ○ Correlation does not imply causation — both are caused by hot weather
- ○ The data must be wrong
- ○ There is no correlation

Question 7



Picture each scatter plot.

Which pair would make a scatter plot with no pattern (no correlation)?

- ○ Hours studied vs. test score
- ○ Height vs. weight
- ○ Shoe size vs. IQ score
- ○ Exercise vs. weight loss

Question 8

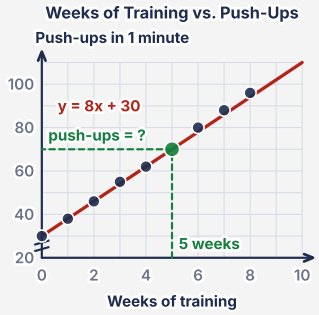


A trend line is described as "strong and positive." What does each word mean?

- ○ Strong = steep slope · Positive = y is positive
- ○ Strong = points cluster tightly near the line · Positive = trend goes upward
- ○ Strong = many data points · Positive = all y-values are positive
- ○ Strong = the line is long · Positive = x values increase

Interpret in context: slope = "per unit increase in x" · y-intercept = "when x = 0"

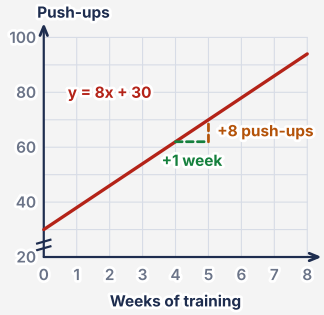
Question 1



The model is y = 8x + 30 (x = training weeks, y = max push-ups in 1 min). Find the green dot: how many push-ups after **5 weeks**?

- ○ 38 push-ups
- ○ 70 push-ups
- ○ 55 push-ups
- ○ 80 push-ups

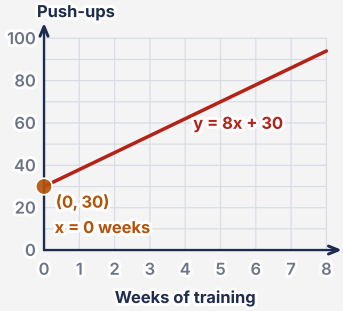
Question 2



In y = 8x + 30 (training weeks vs. push-ups), interpret the slope of 8.

- ○ The athlete starts at 8 push-ups
- ○ For each additional week of training, push-up count increases by about 8
- ○ Training takes 8 weeks total
- ○ The slope means 8 athletes were tested

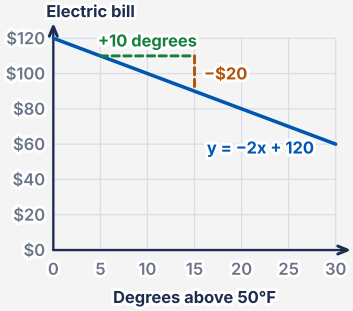
Question 3



Interpret the y-intercept of 30 in the same model (y = 8x + 30).

- ○ 30 is the maximum push-up count
- ○ At 0 weeks of training, a person can do about 30 push-ups
- ○ It takes 30 weeks to improve
- ○ 30 people were surveyed

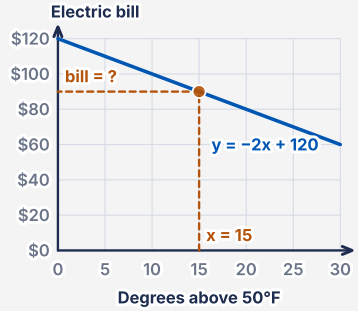
Question 4



Temperature vs. electric heating bills: y = −2x + 120 (x = °F above 50, y = bill in $). What does slope −2 mean?

- ○ The bill increases $2 per degree above 50°F
- ○ For each degree above 50°F, the bill decreases by $2
- ○ The starting bill is −$2
- ○ The bill is always negative

Question 5



Using y = −2x + 120 from Q4: what is the predicted bill at 15 degrees above 50°F?

- ○ $90
- ○ $105
- ○ $150
- ○ $120

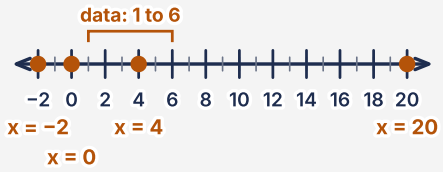
Question 6



Model: y = 4x + 8 (x = hours tutoring, y = points gained). The data was collected for x = 1 to x = 6. Is x = 10 inside or outside the data? Is that prediction safe?

- ○ Inside the data — safe
- ○ Outside the data — less safe
- ○ Correlation — measuring association
- ○ Causation — tutoring causes gains

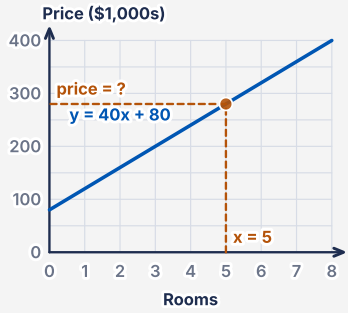
Question 7



Using y = 4x + 8 from Q6, which is the most reliable prediction?

- ○ x = 0 hours (no data there)
- ○ x = 4 hours (within the data range)
- ○ x = 20 hours (beyond data)
- ○ x = −2 hours (impossible)

Question 8



A researcher finds y = 40x + 80 models home price. x = number of rooms. y = price in $1,000s. What is the predicted price for a home with 5 rooms (x = 5)?

- ○ $280,000
- ○ $80,000
- ○ $200,000
- ○ $280

📋 Practice 2 Two-Way Table

120 8th grade students surveyed on sport preference split by **lunch period**:

	Basketball 🏀	Soccer ⚽	Other 🏃	Total
1st Lunch	24	**18**	18	**60**
2nd Lunch	30	12	18	**60**
Total	**54**	**30**	**36**	**120**

Question 1

	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

Find where the 1st Lunch row and the Soccer column cross.

What is the joint frequency of 1st lunch students who prefer Soccer?

- ○ 30
- ○ 18
- ○ 12
- ○ 60

Question 2

	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

Marginal = the total at the edge of the Basketball column.

What is the marginal frequency for Basketball?

- ○ 24
- ○ 30
- ○ 54
- ○ 120

Question 3

	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

1st Lunch: $\frac{24}{60} = ?$

What percent of 1st lunch students prefer Basketball?

- ○ 40%
- ○ 45%
- ○ 20%
- ○ 50%

Question 4

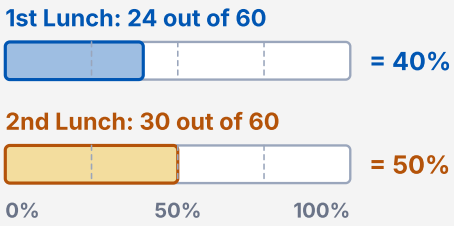
	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

2nd Lunch: $\frac{30}{60} = ?$

What percent of 2nd lunch students prefer Basketball?

- ○ 40%
- ○ 50%
- ○ 45%
- ○ 25%

Question 5



Based on the basketball percentages (40% vs 50%), is there an association?

- ○ No — the totals are both 60
- ○ Yes — 2nd lunch students prefer basketball more (50% vs 40%)
- ○ No — 40 and 50 are close enough
- ○ Yes — but only if we add more data

Question 6

	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

Soccer: $\frac{30}{120} = ?$

What is the relative frequency of students who prefer Soccer (out of 120)?

- ○ 25%
- ○ 30%
- ○ 15%
- ○ 18%

Question 7

	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

Which cell is highlighted in green? What does that value represent?

- 1st Lunch × Basketball = 24 (most popular combo for 1st lunch)
- 1st Lunch × Soccer = 18 (joint frequency)
- Soccer column total = 30
- Grand total = 120

Question 8

	Basket-ball	Soccer	Other	Total
1st Lunch	24	18	18	60
2nd Lunch	30	12	18	60
Total	54	30	36	120

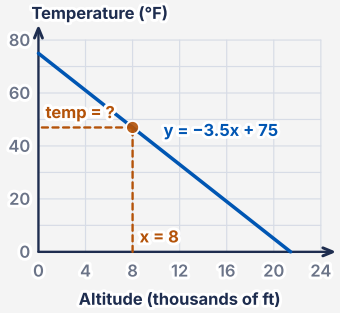
Everyone who picked Other, out of all 120 students.

If a student is chosen randomly from the 120 surveyed, what is the probability they prefer "Other" sports?

- $^{36}/_{90}$ = 40%
- $^{36}/_{60}$ = 60%
- $^{36}/_{120}$ = 30%
- $^{18}/_{120}$ = 15%

Extended thinking: These questions require connecting multiple concepts. Take your time and show your reasoning.

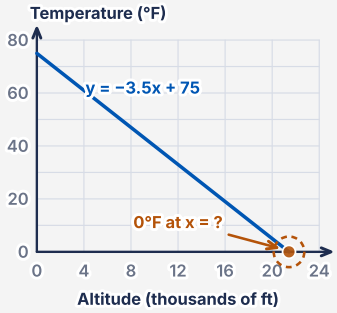
Question 1



The line y = −3.5x + 75 models temperature (°F) at altitude x (in 1,000s of feet). At 8,000 ft (x = 8), predict the temperature.

- 47°F
- 75°F
- 28°F
- 47°C

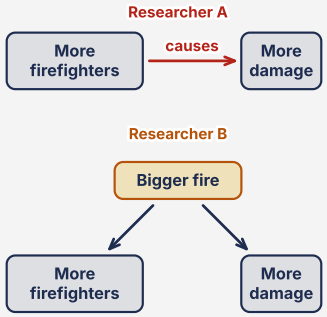
Question 2



Using the same model (y = −3.5x + 75): at what altitude does the model predict 0°F? Round to the nearest 1,000 ft.

- x ≈ 21,000 ft
- x ≈ 75,000 ft
- x ≈ 15,000 ft
- x ≈ 10,000 ft

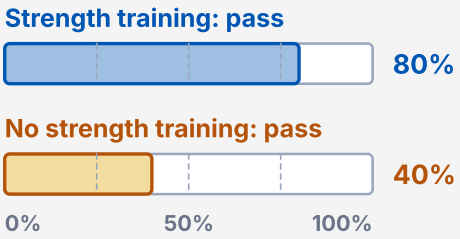
Question 3



Two researchers study "firefighters at scene vs. fire damage." They find a strong positive correlation. Researcher A says: "More firefighters cause MORE damage." Researcher B says: "More firefighters are sent to BIGGER fires, which also cause more damage." Who is correct?

- ○ Researcher A — the data shows clear causation
- ○ Researcher B — larger fires need more firefighters AND cause more damage (lurking variable)
- ○ Both are wrong — there is no correlation
- ○ Both are correct — it depends on the day

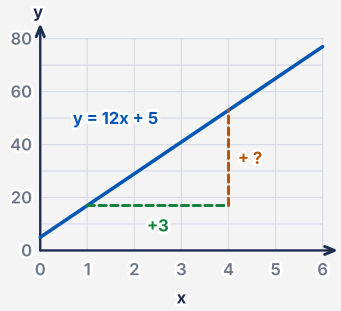
Question 4



A two-way table shows: 80% of athletes who do strength training pass fitness tests, vs. 40% of athletes who don't. The difference is 40 percentage points. What does this suggest?

- ○ There is no association
- ○ There is a strong association between strength training and passing fitness tests
- ○ The sample must be biased
- ○ The difference is too small to matter

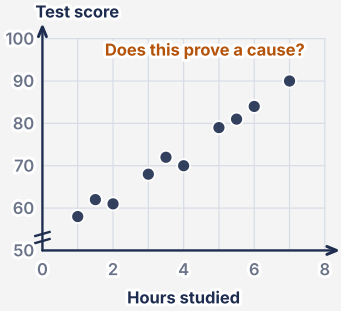
Question 5



Line of best fit: y = 12x + 5. If x increases by 3 units, by how much does y increase?

- ○ 12
- ○ 15
- ○ 36
- ○ 5

Question 6



Can we use a scatterplot to PROVE that studying more CAUSES higher test scores? Explain.

- ○ Yes — a strong positive correlation proves causation
- ○ No — correlation does not prove causation; other factors may be involved
- ○ Yes — if the trend line is very accurate
- ○ No — only if there are more than 100 data points

📝 Worked Examples

Study these step-by-step solutions before attempting the problems above.

Two-way table: 45 boys like pizza out of 120 total

Step 1: Joint relative freq

```
45 ÷ 120 = 0.375 = 37.5%
```

	Like pizza	Do not like	Total
Boys	45	15	60
Girls			60
Total			120

Joint: $\frac{45}{120} = 37.5\%$

Step 2: Conditional: of 60 boys

```
45 ÷ 60 = 0.75 = 75% of boys like pizza
```

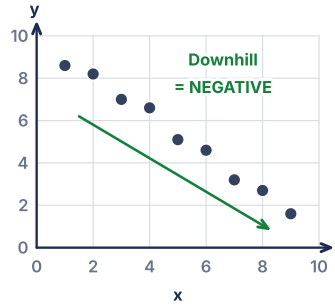
	Like pizza	Do not like	Total
Boys	45	15	60
Girls			60
Total			120

Of boys: $\frac{45}{60}$ = 75%

🔍 Common Mistakes

Can you spot what went wrong?

Correlation: points go up-left to down-right



✗ **Student:** Student says: positive correlation

✓ **Fix:** WRONG! Downhill = NEGATIVE correlation. Uphill = positive.

💡 *Think of it like a hill: going up = positive, going down = negative.*

⚡ Key Tips for This Unit

✅ Positive: both increase. Negative: one up, other down. None: random.

✅ Line of best fit: approx equal points above and below.

✅ Outlier: far from the pattern. Can distort the trend line.

✅ Joint RF = cell÷grand total. Conditional RF = cell÷row or column total.